10.10.2 Bonding in some Homonuclear di-atomic molecules:

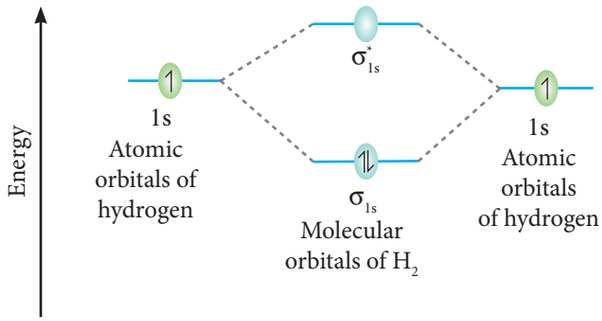


Fig 10.30 MO Diagram for H_2 molecule

Molecular orbital diagram of hydrogen molecule (H_2)

Electronic configuration of H atom is **$1s^1$**

Electronic configuration of H_2 molecule σ_{1s}^2

Bond order = $\frac{N_b - N_a}{2} = \frac{2-0}{2} = 1$

Molecule has *no unpaired electrons.* Hence, it is **diamagnetic**.

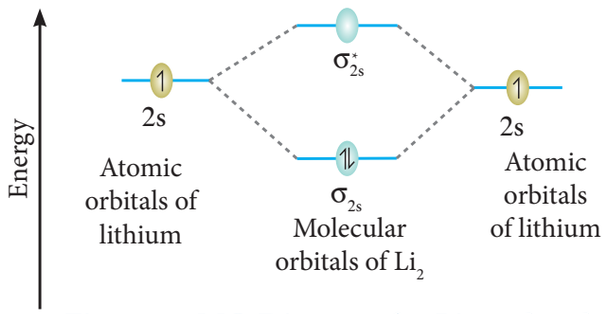


Fig 10.31 MO Diagram for Li_2 molecule

Molecular orbital diagram of lithium molecule (Li_2)

Electronic configuration of Li atom is **$1s^2\,2s^1$**

Electronic configuration of Li_2 molecule $\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2$

Bond order = $\frac{N_b - N_a}{2} = \frac{4-2}{2} = 1$

Molecule has *no unpaired electrons.* Hence it is **diamagnetic**.

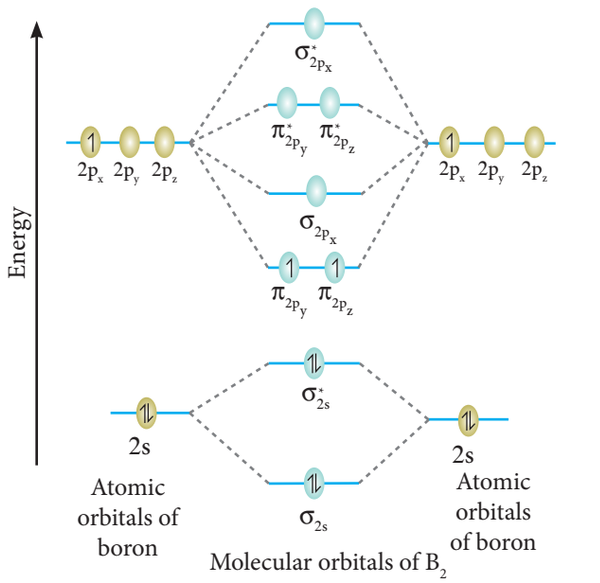


Fig 10.32 MO Diagram for B_2 molecule

Molecular orbital diagram of boron molecule (B_2)

Electronic configuration of B atom is **$1s^2\,2s^2\,2p^1$**

Electronic configuration of B_2 molecule $\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}, \pi^1_{2p_y}, \pi^1_{2p_z}$

Bond order = $\frac{N_b - N_a}{2} = \frac{6-4}{2} = 1$

Molecule has *two unpaired electrons. Hence* it is **paramagnetic.**

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